All modules — derivation summary

Each derivation: the equation it starts from and the result it reaches

Module I — Crystal structure & X-rays

Only 1, 2, 3, 4 and 6-fold rotation axes are allowed

5 marks

A, B, C, D: lattice points on one row, T apart. An n-fold axis turns the lattice into itself by $\theta = 2\pi/n$. B', C': where A and D land after turning about B and C. m: a whole number.

STARTING FROM

$$\theta = \frac{2\pi}{n}$$

FINAL RESULT

$$n = 1, 2, 3, 4, 6$$

Interplanar spacing d_{hkl}

5 marks

The plane (hkl) nearest the origin cuts the axes at $OA = a/h$, $OB = b/k$, $OC = c/l$. $ON = d$ is the perpendicular from the origin to the plane; α, β, γ are its angles with the axes. The axes are at right angles.

STARTING FROM

$$OA = \frac{a}{h}, \quad OB = \frac{b}{k}, \quad OC = \frac{c}{l}$$

FINAL RESULT

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Packing fraction of BCC = 0.68

4 marks

Cube side a, atomic radius r. In BCC the atoms touch along the body diagonal.

STARTING FROM

$$n = 8 \times \tfrac{1}{8} + 1$$

FINAL RESULT

$$\mathrm{PF} = \frac{\sqrt{3}\,\pi}{8} \approx 0.68$$

Packing fraction of FCC = 0.74

4 marks

Cube side a, atomic radius r. Packing fraction = volume filled by atoms in one cell ÷ volume of the cell.

STARTING FROM

$$n = 8 \times \tfrac{1}{8} + 6 \times \tfrac{1}{2}$$

FINAL RESULT

$$\mathrm{PF} = \frac{\pi}{3\sqrt{2}} \approx 0.74$$

Ideal HCP: $c/a = \sqrt{8/3} = 1.633$

5 marks

P, Q, R: three touching atoms of layer A, side a. T: the layer-B atom resting in their hollow. G: centre of triangle PQR, right below T. M: midpoint of PQ. Layers A and B are $c/2$ apart.

STARTING FROM

$$PM = \frac{a}{2}$$

FINAL RESULT

$$\frac{c}{a} = \sqrt{\frac{8}{3}} \approx 1.633$$

Bragg's law, $2d\sin\theta = n\lambda$

4–5 marks

Planes (hkl) are a distance d apart. θ is the glancing angle, measured from the plane. Δ is the path difference between the two reflected rays, n the order, λ the wavelength.

STARTING FROM

$$\Delta = PB + BQ$$

FINAL RESULT

$$2d\sin\theta = n\lambda$$

Laue's equations, and Bragg's law from them

5–8 marks

A, B: identical scattering atoms, $\mathbf{r} = \overrightarrow{AB}$. $\hat{\mathbf{n}}_1$, $\hat{\mathbf{n}}_2$: unit vectors along the incident and scattered beams; the angle between them is 2θ. $\mathbf{N} = \hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$. α, β, γ: angles of $\mathbf{N}$ with the axes $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

STARTING FROM

$$\Delta = AM - AL$$

FINAL RESULT

$$2d\sin\theta = n\lambda$$

The reciprocal of a BCC lattice is FCC

5 marks

a: cube edge. $\hat{i}, \hat{j}, \hat{k}$: unit vectors along the cube edges. $\mathbf{a}', \mathbf{b}', \mathbf{c}'$: primitive vectors of BCC. Reciprocal vectors: $\mathbf{a}^* = 2\pi\,\dfrac{\mathbf{b}' \times \mathbf{c}'}{\mathbf{a}' \cdot (\mathbf{b}' \times \mathbf{c}')}$ and cyclically.

STARTING FROM

$$\mathbf{a}' = \tfrac{a}{2}(\hat{i} + \hat{j} - \hat{k})$$

FINAL RESULT

$$\mathbf{a}^* = \frac{2\pi}{a}(\hat{i} + \hat{j})$$

Bragg's law in the reciprocal lattice

5 marks

$\mathbf{k}, \mathbf{k}'$: incident and diffracted wave vectors, $|\mathbf{k}| = |\mathbf{k}'| = 2\pi/\lambda$. $\mathbf{G}$: a reciprocal lattice vector, normal to the planes (hkl), $|\mathbf{G}| = 2\pi n/d$. θ: glancing angle.

STARTING FROM

$$\mathbf{a} \cdot (\mathbf{k}' - \mathbf{k}) = -2\pi n h$$

FINAL RESULT

$$2d\sin\theta = n\lambda$$

Module II — Theory of solids

Drude conductivity $\sigma = ne^2\tau/m$

4–5 marks

n: free electrons per unit volume. τ: mean free time between collisions (relaxation time). v_d: drift velocity. $\mathcal{E}$: applied field. J: current density.

STARTING FROM

$$m\frac{dv}{dt} = -e\mathcal{E}$$

FINAL RESULT

$$\sigma = \frac{ne^2\tau}{m}$$

Free electrons in a 1-D box: E_n and E_F

5 marks

A line of length L with infinite walls at $x = 0$ and $x = L$; $V = 0$ inside. E_n: energy of level n. N: number of electrons. n_F: the highest filled level.

STARTING FROM

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

FINAL RESULT

$$E_F = \frac{h^2 N^2}{32mL^2}$$

Fermi energy of a 3-D electron gas

5–8 marks

Cube of side L, volume $V = L^3$, periodic boundary conditions. Allowed k_x, k_y, k_z are multiples of $2\pi/L$. N electrons, density $n = N/V$. k_F: radius of the filled sphere at 0 K.

STARTING FROM

$$k_x = \frac{2\pi n_x}{L}$$

FINAL RESULT

$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}$$

Density of states $D(E)$ in 3-D

5 marks

$N(E)$: number of electron states with energy less than E. $D(E) = dN/dE$: number of states per unit energy range. Cube of volume V.

STARTING FROM

$$N(E) = \frac{V}{3\pi^2}k^3$$

FINAL RESULT

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

Average energy at 0 K is $\frac{3}{5}E_F$

4 marks

$D(E) = CE^{1/2}$ with $C = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2}$. At 0 K every state below E_F is filled and every state above is empty.

STARTING FROM

$$E_{\text{tot}} = \int_0^{E_F} E\,D(E)\,dE$$

FINAL RESULT

$$\bar{E}_0 = \frac{3}{5}E_F$$

Electronic specific heat: only a fraction T/T_F can be excited

5 marks

$T_F = E_F/k_B$ (Fermi temperature). Classically every electron would take $\frac{3}{2}k_BT$.

STARTING FROM

$$\left\langle \tfrac{1}{2}mv_x^2 \right\rangle = \frac{\int \frac{1}{2}mv_x^2\, e^{-mv_x^2/2k_BT}\,dv_x}{\int e^{-mv_x^2/2k_BT}\,dv_x} = \tfrac{1}{2}k_BT$$

FINAL RESULT

$$C_v = \frac{3}{2}Nk_B\frac{T}{T_F}$$

Bloch theorem: $\psi(x+a) = e^{ika}\psi(x)$

5–8 marks

$V(x+a) = V(x)$: a periodic potential with lattice constant a. $f(x)$, $g(x)$: two real independent solutions of the Schrödinger equation. λ: a constant.

STARTING FROM

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0$$

FINAL RESULT

$$\psi(x) = e^{ikx}u_k(x)$$

The Kronig–Penney model

8 marks

Wells of width a ($V = 0$) between barriers of width b and height V_0; period $a + b$. $E < V_0$. $\alpha^2 = 2mE/\hbar^2$, $\beta^2 = 2m(V_0 - E)/\hbar^2$.

STARTING FROM

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad (0 < x < a)$$

FINAL RESULT

$$P\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka$$

Velocity of a Bloch electron: $v = \frac{1}{\hbar}\frac{dE}{dk}$

3 marks

An electron in a band is a wave packet; it moves with the group velocity $v_g = d\omega/dk$. $E = \hbar\omega$.

STARTING FROM

$$\cos(kx-\omega t)+\cos\big((k+\delta k)x-(\omega+\delta\omega)t\big)=2\cos\tfrac{1}{2}(\delta k\,x-\delta\omega\,t)\,\cos(\text{fast wave})$$

FINAL RESULT

$$v=\frac{1}{\hbar}\frac{dE}{dk}$$

Effective mass $m^*=\hbar^2/(d^2E/dk^2)$

5 marks

$\mathcal{E}$: applied field. $v=\frac{1}{\hbar}\frac{dE}{dk}$ (group velocity). The electron's charge is taken as e in magnitude; only the form of the result matters here.

STARTING FROM

$$dE=e\mathcal{E}\,v\,dt$$

FINAL RESULT

$$m^*=\frac{\hbar^2}{d^2E/dk^2}$$

Module III — Semiconductor properties

Conductivity of a semiconductor: $\sigma=e(n\mu_n+p\mu_p)$

4 marks

v_d: drift velocity. μ: mobility, the drift velocity per unit field. n, p: electron and hole concentrations; μ_n, μ_p their mobilities. $\mathcal{E}$: applied field.

STARTING FROM

$$v_d=\mu\mathcal{E}$$

FINAL RESULT

$$\sigma_i=en_i(\mu_n+\mu_p)$$

$\ln\sigma$ against $1/T$ measures the band gap

4 marks

Mobilities taken as independent of temperature. n_i from the law of mass action.

STARTING FROM

$$\sigma=en_i(\mu_n+\mu_p)$$

FINAL RESULT

$$\ln\sigma=-\frac{E_g}{2k}\cdot\frac{1}{T}+\frac{3}{2}\ln T+\text{const}$$

Electron concentration in the conduction band

8 marks

$D(E)$: density of states per unit volume. $f(E)$: Fermi function. m_n^*: effective mass of electrons in the conduction band. E_c: bottom of the conduction band. Kinetic energy of an electron at E is $E-E_c$.

STARTING FROM

$$dn=D(E)f(E)\,dE$$

FINAL RESULT

$$n=N_c\,e^{-(E_c-E_F)/kT}$$

$$N_c=2\left(\frac{2\pi m_n^*kT}{h^2}\right)^{3/2}$$

Hole concentration in the valence band

5 marks

$1-f(E)$: probability that a state is empty, i.e. holds a hole. m_p^*: hole effective mass. E_v: top of the valence band; a hole at E has kinetic energy $E_v - E$.

STARTING FROM

$$dp = D(E)\,[1-f(E)]\,dE$$

FINAL RESULT

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

Fermi level of an intrinsic semiconductor

5 marks

Intrinsic: $n = p = n_i$. N_c, N_v as in the two previous derivations.

STARTING FROM

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

FINAL RESULT

$$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4}\ln\frac{m_p^*}{m_n^*}$$

Law of mass action and the intrinsic concentration n_i

5 marks

$E_g = E_c - E_v$: band gap. n, p from the two concentration derivations.

STARTING FROM

$$np = N_c N_v e^{-(E_c - E_F)/kT} e^{-(E_F - E_v)/kT}$$

FINAL RESULT

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

n-type: Fermi level and electron concentration

8 marks

N_d: donors per unit volume at level E_d, just below E_c. N_d^+: ionised donors. $\Delta E = E_c - E_d$: ionisation energy of a donor. Moderate temperature: holes are negligible, and E_F is a few kT from both E_c and E_d.

STARTING FROM

$$n = N_d^+ + p \approx N_d^+$$

FINAL RESULT

$$n = (N_d N_c)^{1/2} e^{-\Delta E/2kT}$$

Module IV — Magnetic & dielectric properties

Langevin theory of diamagnetism

8 marks

An electron (charge $-e$, mass m) goes round a nucleus $+Ze$ in a circle of radius ρ with angular frequency ω_0. A field B is applied at right angles to the orbit. N atoms per unit volume, Z electrons each.

STARTING FROM

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2}$$

FINAL RESULT

$$\chi_{\text{dia}} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2\rangle$$

Langevin theory of paramagnetism and the Curie law

8 marks

N atoms per unit volume, each with a permanent moment μ, free to turn, not interacting. θ: angle between a moment and $\mathbf{B}$. k: Boltzmann's constant.

STARTING FROM

$$\tau = \mu B\sin\theta$$

FINAL RESULT

$$\chi_{\text{para}} = \frac{\mu_0 N\mu^2}{3kT} = \frac{C}{T}$$

Quantum theory of paramagnetism (small-field result)

5 marks

Moment $\mu = g\mu_B J$ with J the total angular momentum, g the Landé factor, $\mu_B = e\hbar/2m$ the Bohr magneton. Only $2J+1$ orientations are allowed: components $-g\mu_B m_J$, $m_J = -J,\ldots,J$.

STARTING FROM

$$\mu = -\frac{\mu_B}{\hbar}(\mathbf{L}+2\mathbf{S})$$

FINAL RESULT

$$\chi = \frac{\mu_0 N g^2 J(J+1)\mu_B^2}{3kT} = \frac{\mu_0 N p_{\text{eff}}^2\mu_B^2}{3kT}$$

Weiss theory: the Curie–Weiss law

8 marks

Weiss: (i) a ferromagnet is made of domains, each spontaneously magnetised; (ii) inside a domain an exchange (molecular) field $B_E = \lambda M$ acts, λ the Weiss constant. Paramagnetic region $T > T_c$.

STARTING FROM

$$B_{\text{eff}} = B + \lambda M$$

FINAL RESULT

$$\chi = \frac{C}{T-T_c}, \quad T_c = \frac{\lambda C}{\mu_0}$$

Néel's theory of antiferromagnetism above T_N

5 marks

Two interpenetrating sublattices A and B, equal moments, N atoms each per unit volume. Molecular field constants: α for A–A (and B–B), β for A–B. Both interactions are negative (antiparallel).

STARTING FROM

$$B_A = B - \alpha M_A - \beta M_B$$

FINAL RESULT

$$\chi = \frac{2C}{T+\theta}$$

The Lorentz local field $E_{\text{loc}} = E + P/3\varepsilon_0$

6 marks

$\boldsymbol{E}$: macroscopic field in the dielectric, $\boldsymbol{P}$: polarisation. Carve an imaginary sphere of radius $\boldsymbol{a}$ round the atom. $\boldsymbol{\theta}$: angle from the field direction.

STARTING FROM

$$E_{\text{loc}} = E_1 + E_2 + E_3 + E_4$$

FINAL RESULT

$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

Clausius–Mossotti relation

5 marks

$\boldsymbol{\alpha}$: polarisability of one atom, $p = \alpha E_{\text{loc}}$. N atoms per unit volume. ε_r: dielectric constant, $D = \varepsilon_0 \varepsilon_r E = \varepsilon_0 E + P$.

STARTING FROM

$$P = N\alpha E_{\text{loc}}$$

FINAL RESULT

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$$

Electronic polarisability (classical oscillator)

5 marks

An electron bound harmonically to its atom: force constant β, natural frequency $\omega_0 = \sqrt{\beta/m}$. x: its displacement in the local field.

STARTING FROM

$$m\frac{d^2x}{dt^2} = -\beta x \;\Rightarrow\; \omega_0 = \sqrt{\frac{\beta}{m}}$$

FINAL RESULT

$$\alpha_e = \frac{e^2}{m(\omega_0^2 - \omega^2)}$$

Ionic polarisability

3 marks

Positive and negative ions displace relative to each other by Δx; β: force constant of the bond.

STARTING FROM

$$\beta \Delta x = eE$$

FINAL RESULT

$$\alpha_i = \frac{e^2}{\beta}$$

Dipolar (orientational) polarisability $\alpha_d = p^2/3kT$

5 marks

Molecules with a permanent dipole moment p (e.g. $\mathrm{H_2O}$), N per unit volume. θ: angle between a dipole and $\mathbf{E}$.

STARTING FROM

$$\tau = pE\sin\theta$$

FINAL RESULT

$$\alpha_d = \frac{p^2}{3kT}$$

Ferroelectricity: the polarisation catastrophe and $\varepsilon_r \propto 1/(T - T_c)$

5 marks

Clausius–Mossotti with electronic plus ionic polarisabilities $\boldsymbol{\alpha_j}$ of the ions, $\boldsymbol{N_j}$ per unit volume; local field $\boldsymbol{E + P/3\varepsilon_0}$ at every ion.

STARTING FROM

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = S, \quad S = \frac{1}{3\varepsilon_0}\sum_j N_j \alpha_j$$

FINAL RESULT

$$\varepsilon_r = \frac{3\xi}{T - T_c}$$